

```

void setup()
{
    Serial.begin(115200);
    while (!Serial)
        delay(10); // will pause Zero, Leonardo, etc until serial console opens

    Serial.println("Adafruit MPU6050 test!");

    // Try to initialize!
    if (!mpu.begin()) {
        Serial.println("Failed to find MPU6050 chip");
        while (1) {
            delay(10);
        }
    }
    Serial.println("MPU6050 Found!");

    //setupt motion detection
    mpu.setHighPassFilter(MPU6050_HIGHPASS_0_63_HZ);
    mpu.setMotionDetectionThreshold(1);
    mpu.setMotionDetectionDuration(20);
    mpu.setInterruptPinLatch(true); // Keep it latched. Will turn off when reinitialize
    mpu.setInterruptPinPolarity(true);
    mpu.setMotionInterrupt(true);

    Serial.println("");
    delay(100);

    Blynk.begin(auth, ssid, pass);
    timer.setInterval(100L, sendSensor);
}

void loop()
{
    Blynk.run();
    timer.run();
}

```

Fig. 8: Code snippet for setup function

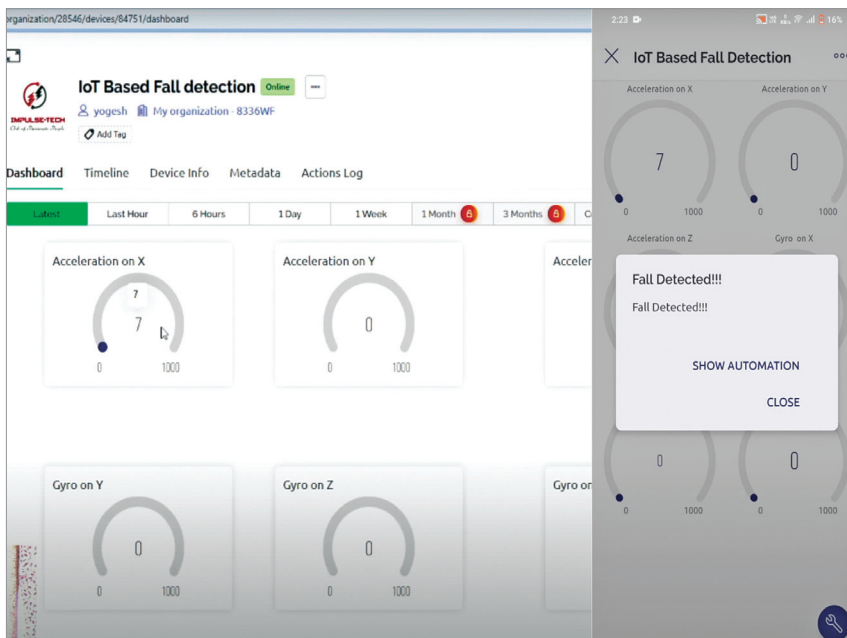


Fig. 9: Blynk showing the alert and data

7 shows the code snippet for sensor data capturing.

The 'sendSensor()' function reads data from the MPU6050 sensor and sends it to the Blynk app. It checks if a motion interrupt has occurred, reads accelerometer and gyroscope data, and sends this data to the Blynk app using virtual pins (V0 to V5). A delay is introduced to limit the data sending rate. Fig. 8 shows the code snippet for setup function.

The 'setup()' function initialises serial communication, sets up the MPU6050 sensor, configures motion detection settings, connects to Wi-Fi and the Blynk server using provided authentication credentials. The 'sendSensor()' function is scheduled to run at regular intervals. In the 'loop()' function, 'Blynk.run()' and 'timer.run()' are called to handle Blynk communications and timer-based tasks, respectively.

This code is for an IoT-based fall detection system that uses an ESP32, MPU6050 sensor, and the Blynk platform to monitor and transmit motion and orientation data, making it suitable for fall detection applications.

Construction and testing

Connect the sensor according to the circuit diagram shown in Fig. 2, upload the code to the board, and power the device with a USB cable or 5V battery. Wait for the device to connect with Wi-Fi. Now make the device fall or change the acceleration of the device rapidly; it detects and displays it on the Blynk dashboard and shows the alert on the phone as shown in Fig. 9. **EFY**



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